How Much Adaptation Is Enough?
Drift Geometry and Minimal Calibration in Brain-to-Text BCI

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Abstract

Brain-to-text BCI decoders must remain usable as neural signals change across recording sessions. We audit how much input adaptation is needed when the official T15 brain-to-text GRU is evaluated across sessions. Native day-specific input layers reproduce low validation phoneme error (9.06% PER), while forcing sessions through non-native input layers increases PER to 22.67–34.43%. Unsupervised moment corrections fail to recover this loss. A supervised input-layer adapter trained from 20 labeled calibration trials improves the fixed-middle stress setting from 22.67% to 18.59% PER, but still leaves substantial degradation. The strongest practical result is source-layer selection: using a short unlabeled beginning-of-day window to choose a past input layer reaches 12.74% PER at $K = 20$, while the immediately previous session is even stronger at 12.21%. Secondary geometry-only analyses of T12 diagnostic blocks and inner-speech behavior blocks show similar state structure beyond the main T15 decoder experiment. These results suggest that recency should be the default adaptation prior, while drift geometry is a useful signal for deciding when that default may need to be overridden.

1 Introduction

High-performance speech BCIs commonly include session-specific adaptation machinery because intracortical neural features drift across days. Long-term stability has long been recognized as a central challenge for practical BCI deployment [Vaadia and Birbaumer, 2009]. In this work, "drift" means that the neural feature distribution recorded on a new day no longer matches the distribution seen by the decoder during earlier sessions. A practical BCI therefore needs a way to decide whether yesterday's input mapping is still usable, whether an older mapping is a better match, or whether a short calibration step is needed. This raises a practical question: before retraining a decoder or collecting a long supervised calibration set, how much adaptation is enough? Our main practical finding is that cross-session BCI drift is often better handled by choosing the right input state than by retraining the decoder.

We examine this question in the official T15 brain-to-text GRU baseline from a rapidly calibrating speech neuroprosthesis study [Card et al., 2024]. The model contains learned day-specific input layers, followed by a GRU and output head. Intuitively, each input layer acts like a small session-specific adapter that maps that day's neural features into the representation expected by the shared GRU decoder. This structure creates a useful stress test: if a target session is decoded through the wrong input layer, performance measures the decoder-facing cost of cross-session input mismatch. We then ask whether simple feature corrections, small learned adapters, or geometry-guided source selection can recover the lost performance.

The contribution is not a new speech decoder. Instead, we provide a decoder-facing adaptation audit:

- Wrong input states break the decoder: native-day PER is 9.06%, while fixed non-native input layers increase PER to 22.67–34.43%.
- Simple mean/scale correction is insufficient; input-layer calibration with $K = 20$ improves PER to 18.59% but remains far from native-day.
- Previous-session reuse is the strongest simple policy at $K = 20$ (12.21% PER), while geometry selection is close (12.74%) and reveals older-state candidates.
- Geometry is a retrieval signal, not a standalone policy: at $K = 20$ it selects older states in 11 sessions, but wins only 3 times, motivating a confidence gate.

We do not claim that neural drift geometry is new. Rather, we show that in a deployed-style brain-to-text decoder with learned day-specific input layers, short-window geometry is actionable for choosing among existing input-layer states and for framing recency as an explicit adaptation prior. Figure 1 summarizes this adaptation-ladder view.

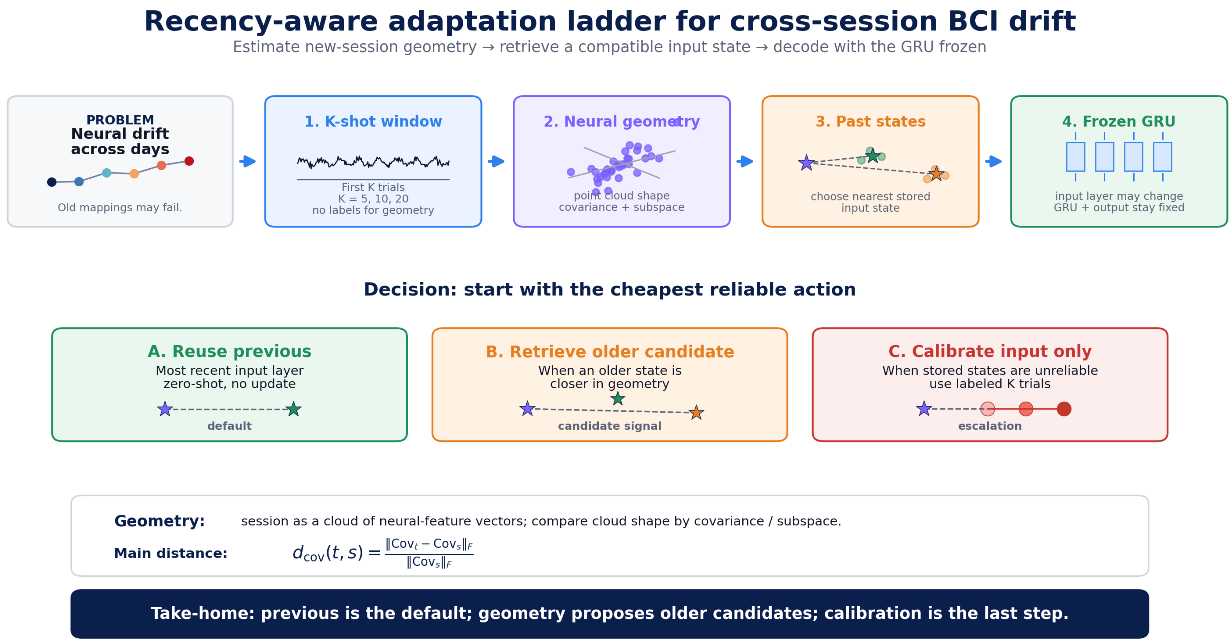

Figure 1: Recency-aware input-state selection for cross-session BCI drift. For a new target session, the first K trials are used to estimate neural feature geometry. Each session is represented as a cloud of neural feature vectors, summarized by covariance and subspace statistics. The target geometry is compared to a library of past input states. The system first reuses the previous input layer when it is recent and plausible; otherwise, geometry can retrieve an older candidate state. If no stored state is reliable, only the input layer is lightly calibrated while the shared GRU decoder remains frozen.

2 Related Work

2.1 Brain-to-Text BCIs

Intracortical communication BCIs have advanced rapidly from handwriting-based brain-to-text decoding [Willett et al., 2021; Fan et al., 2023] to speech neuroprostheses that decode attempted speech into text or spoken output [Willett et al., 2023; Card et al., 2024]. These systems commonly rely on recurrent neural decoders and session-specific adaptation or recalibration to maintain performance as recordings change over time. Recent work also explores transferring pretrained speech representations to brain decoding, for example by adapting Wav2Vec2-style models to neural features [Fiedler et al., 2025]. That direction asks how stronger speech backbones or pretrained acoustic representations can improve neural decoding. Our work is complementary: given an already trained brain-to-text decoder with day-specific input layers, we ask how to keep the decoder usable across sessions using lightweight input-side adaptation rather than retraining the backbone.

2.2 Cross-Session Stability and Neural Geometry

Cross-session stability has been studied through the lens of neural geometry. Manifold-alignment methods can stabilize fixed decoders by aligning low-dimensional neural activity across recording instabilities [Degenhart et al., 2020], and recent dynamics-aware approaches such as NoMAD use temporal structure to perform unsupervised alignment over weeks to months [Karpowicz et al., 2025]. Our analysis uses this geometry perspective in a narrower decoder-facing setting: we measure whether short-window geometry can guide selection among existing day-specific input-layer states.

2.3 Session-Invariant and Continual Adaptation

In speech BCI specifically, a recent preprint proposes adversarial session-invariant learning to reduce session-specific information while preserving phoneme decoding structure [Zhang et al., 2026]. Continual recalibration with pseudo-labels provides a complementary route, using language-model corrections to update communication decoders online [Fan et al., 2023]. Our work does not propose a new invariant decoder or full recalibration method. Instead, it audits low-cost choices available before retraining: moment correction, lightweight input calibration, previous-session reuse, and geometry-guided source-layer retrieval.

3 Data and Decoder

T15. The main experiments use T15 copy-task neural features from the official brain-to-text dataset [Card et al., 2024]. We evaluate 41 validation sessions with ground-truth phoneme labels. Neural inputs are passed through the official GRU checkpoint and decoded greedily at the phoneme level. During evaluation, repeated symbols and blank tokens are removed using the standard CTC collapse rule before computing PER. This collapse step is only part of decoding/evaluation; learned calibration is trained separately using CTC loss.

Official baseline. The decoder contains learned day-specific input layers, a GRU backbone, and an output layer over phoneme classes. In the native-day condition, each session uses its own input layer. In cross-day stress tests, target sessions use a fixed or selected non-native source input layer while the rest of the decoder remains unchanged.

Evaluation splits. The main T15 decoder-facing experiments use the validation/test sessions provided with the official dataset. In this split, sessions contain a limited number of held-out trials, so our beginning-of-day experiments treat $K \leq 20$ as the main minimal-calibration regime: the first K trials are used for geometry estimation or supervised input calibration, and all reported PER is computed only on the remaining trials. Larger windows, such as $K = 40$, are useful as exploratory feasibility checks but consume a large fraction of this split and are therefore not used as the main claim. T12 and inner-speech experiments are used only as geometry-only supporting analyses and are not evaluated with decoder PER.

T12 diagnostic blocks. As a secondary check, we analyze diagnostic blocks from the T12 high-performance speech neuroprosthesis dataset [Willett et al., 2023]. This is a geometry-only analysis of spike-power and threshold-crossing feature variants; we do not reproduce the T12 decoder or report T12 PER.

4 Methods

4.1 Decoder-Facing Metrics

For each T15 validation trial, we compute greedy phoneme predictions from the CTC output stream and then apply CTC collapse before computing phoneme error rate (PER). We report phoneme-weighted PER unless otherwise stated. Quality diagnostics such as blank rate, confidence, and entropy are saved, but the core results use PER.

For a set of trials $\mathcal{D}$, the reported PER is

$$\mathrm{PER}(\mathcal{D}) = \frac{\sum_{i \in \mathcal{D}} \mathrm{edit}(y_i, \hat{y}_i)}{\sum_{i \in \mathcal{D}} |y_i|}.$$

We avoid mixing PER with derived recovery percentages in the main result tables: lower PER is always better, and improvements are reported as direct PER changes. This keeps the comparison clear when learned-adapter experiments evaluate on remaining validation trials after holding out the first K calibration trials.

4.2 Cross-Day Stress Test

We compare native-day decoding against three fixed non-native source layers: early (t15.2023.08.13), middle (t15.2023.11.26), and late (t15.2025.04.13). This tests whether the decoder degradation is tied to one bad source layer or is a general consequence of input-layer mismatch.

4.3 Adaptation Ladder

Using the middle source layer, we test no correction, target z-scoring, source moment matching, diagonal affine calibration, and full input-layer calibration. Moment methods use full-session unsupervised feature statistics. Learned adapters use the first K labeled validation trials and are evaluated only on the remaining trials. We use K to simulate a beginning-of-day calibration window: the first K trials are used for adaptation, and the rest of the session is held out for evaluation.

The two unsupervised moment corrections are deliberately simple. Target z-scoring maps x to $(x-\mu_t)/(\sigma_t+\epsilon)$, while source moment matching maps target features into the source marginal statistics:

$$x' = \frac{x - \mu_t}{\sigma_t + \epsilon}(\sigma_s + \epsilon) + \mu_s.$$

These baselines test whether the day-specific input layers can be approximated by channel-wise distribution alignment. Diagonal affine and input-layer calibration are supervised small-calibration probes: the GRU and output head remain frozen, and only a small input transform is learned from the first K validation trials using CTC loss.

4.4 Drift Geometry and Source Selection

Neural feature geometry. We represent each session as a cloud of neural feature vectors in channel space. Each time bin from each trial is one point in this space. The shape of this cloud is summarized using covariance and low-dimensional subspace statistics. For K-shot source selection, the first K unlabeled target trials are used to estimate the target covariance. We then compare this covariance to each past source session using relative Frobenius covariance distance and reuse the input layer of the closest past session.

For each source-target pair, we compute temporal distance, mean and scale shifts, relative covariance shift, CORAL-style distance, and PCA subspace distances. We then select a source input layer by nearest covariance geometry. To approximate deployment, a K-shot variant estimates target geometry from only the first $K \in \{5, 10, 20\}$ unlabeled validation trials and evaluates on the remaining trials. We focus the main claims on $K \leq 20$ because this represents the minimal-calibration regime in the available validation split.

Because temporal recency is an obvious baseline, we also evaluate the immediately previous input layer. Finally, we test a simple recency-aware override: use the previous input layer by default and switch to the geometry-selected older source only if its covariance distance is less than α times the distance to the previous source.

The K-shot source-selection rule is unsupervised with respect to labels. Let $S_t^{(K)}$ denote statistics estimated from the first K target trials, and let S_s denote source-session statistics estimated from past data. The

Table 1: T15 decoder-facing stress test and adaptation ladder. Mean, best-session, and worst-session values are phoneme-weighted PER. Fixed-source and moment-correction rows use the middle source t15.2023.11.26. Learned adapter rows use $K = 20$ calibration trials and are evaluated on remaining trials. Lower PER is better.

Condition	Mean PER ↓	Best session	Worst session
Native-day	9.06%	0.80% (11-19)	23.82% (03-14)
Fixed early	28.81%	7.38% (08-13)	58.28% (03-14)
Fixed middle	22.67%	1.20% (11-19)	56.44% (03-30)
Fixed late	34.43%	16.40% (01-12)	49.04% (07-19)
Moment match	22.73%	1.20% (11-19)	56.90% (03-30)
Diag. affine, $K = 20$	20.21%	1.94% (11-26)	53.26% (03-30)
Input layer, $K = 20$	18.59%	2.46% (11-26)	52.23% (03-30)

geometry-selected source is

$$s^*(t) = \arg\min_{s \in \mathcal{S}_{\mathrm{past}}(t)} d_{\mathrm{cov}}(S_t^{(K)}, S_s),$$

where d_{cov} is relative Frobenius covariance shift. We compare this to a previous-session policy, $s_{\mathrm{prev}}(t)$, because recency is the strongest prior available in a longitudinal BCI.

4.5 Input-State Library Policies

The source-selection experiments can be interpreted as retrieving an input state from a memory of previous sessions. We therefore compare four policies: using only the previous session, selecting geometrically from the last 3 sessions, selecting from the last 5 sessions, and selecting from all past sessions. This ablation asks whether a practical system needs to retain all historical input layers, or whether a bounded recent library is enough under the current geometry metric.

5 Experiments

We organize the empirical study into six experiments, each paired with a table or figure. The T15 experiments are the main decoder-facing results; T12 and inner-speech analyses are supporting geometry-only checks. The experiments follow an adaptation ladder: fixed source → moment correction → learned calibration → source selection → recency/library policies → geometry-only supporting checks.

5.1 Cross-Day Stress Test and Adaptation Ladder

Native-day decoding reaches 9.06% phoneme-weighted PER, confirming that the local evaluation reproduces the expected behavior of the official checkpoint. When all sessions are forced through non-native source input layers, PER rises to 22.67–34.43%, harming 40/41 sessions for each source choice.

The natural first hypothesis is that cross-session mismatch is mostly a mean/variance shift. The data do not support this. Target z-scoring and moment matching to the source distribution remain essentially tied with no correction, and are slightly worse in aggregate. This is consistent with neurobiological evidence that long-term sensorimotor learning involves population-level reorganization rather than only independent single-channel gain changes [Mandelblat-Cerf et al., 2011]. Together, these results suggest that the learned day-specific input layers capture transformations richer than channel-wise mean and variance.

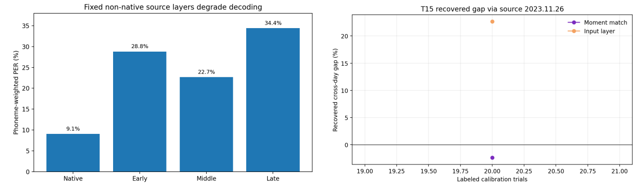

Figure 2: Left: fixed non-native source layers strongly degrade T15 phoneme decoding relative to native-day decoding. Right: small learned calibration improves over the fixed-middle source but remains far from native-day decoding.

Supervised diagonal affine and input-layer calibration improve PER beyond moment matching. With $K = 20$, diagonal affine reduces weighted PER from 21.51% to 20.21%. A longer CUDA input-layer calibration run improves the learned input layer to 18.59%, compared with 22.67% for the fixed-middle source and 9.06% for native-day decoding. A follow-up initialization check suggests that calibration should be treated as residual adaptation rather than reset: on the same $K = 20$ subset, initializing from the previous input state improves 11.58% to 11.03%, whereas fixed-middle initialization remains much weaker at 17.93%. Thus learned input adaptation is a useful rung of the adaptation ladder, but even this stronger small-calibration setting does not replace a well-matched prior input state.

5.2 Beginning-of-Day Source Selection

Geometry-guided source selection performs much better than directly adapting a fixed middle source. In an offline all-candidate setting, choosing the nearest non-native input layer by covariance geometry reaches 11.38% PER. In a stricter past-only setting, it reaches 13.58% PER on 40 sessions, compared with 22.76% for the fixed middle source on the same sessions.

The beginning-of-day version is more realistic. Here, K denotes the first K beginning-of-day trials used for calibration or geometry estimation; evaluation is performed only on the remaining trials. With this protocol, past-only source selection reaches 14.00%, 13.19%, and 12.74% PER at $K = 5, 10, 20$, far below the fixed-middle PER of 22.51%, 22.40%, and 22.30% on the same remaining-trial subsets. However, the immediately previous input layer is even stronger: 13.09%, 12.84%, and 12.21% PER on the same subsets. This prior has a time horizon: previous-source PER is 9.97% when the last available input state is 0–3 days old, 13.75% at 4–14 days, and 24.54% beyond two weeks. Thus previous is a strong default only when it is truly recent. Figure 3 visualizes the pairwise geometry underlying this source-selection problem.

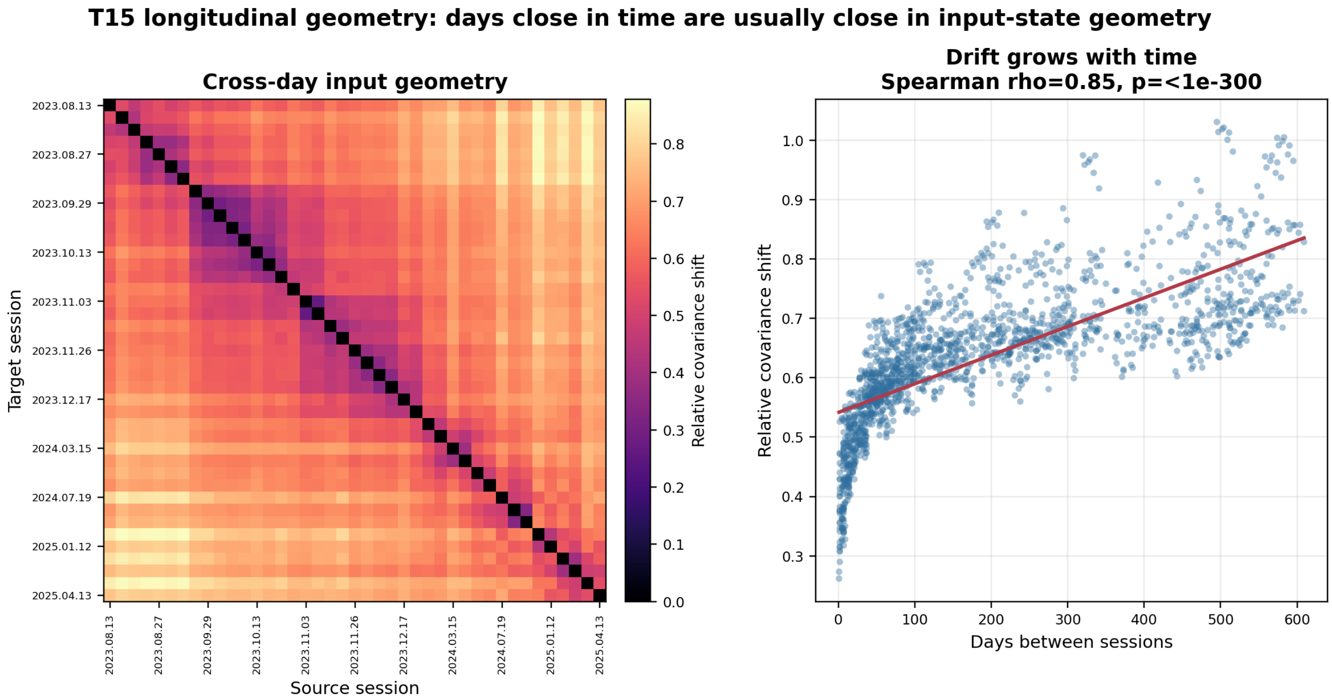

Figure 3: T15 longitudinal geometry map. Left: pairwise relative covariance shift between source and target sessions, ordered by date, shows a diagonal band of nearby sessions with similar input geometry. Right: covariance shift increases with temporal separation across source-target pairs (Spearman $\rho = 0.85$), indicating that session geometry drifts with time while retaining non-monotonic structure.

Table 2: Beginning-of-day source selection on remaining validation trials. Previous-session recency is the strongest simple baseline, while K-shot geometry selection remains close and often selects non-previous sources. Lower PER is better.

Method	$K = 5$	$K = 10$	$K = 20$
Native-day	8.95%	8.88%	8.90%
Fixed middle	22.51%	22.40%	22.30%
Previous source	13.09%	12.84%	12.21%
K-shot geometry source	14.00%	13.19%	12.74%
Best simple override	13.09%	12.90%	12.18%

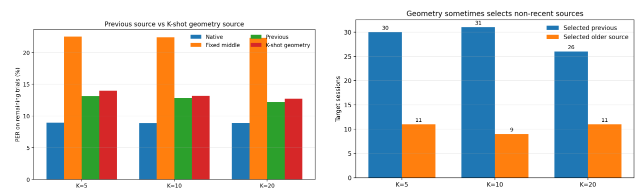

Figure 4: Left: previous-session recency is a strong source-selection baseline and slightly outperforms unconditional K-shot geometry selection. Right: geometry still reveals non-recent source matches, suggesting sessions are not always closest to the immediately previous day.

5.3 Reusable Input-State Libraries

The previous-session input layer is the strongest simple baseline, but it is not always the geometrically nearest source. In T15, K-shot geometry selection chooses the previous session in most targets, but selects an older non-previous input layer in 22–30% of cases. Depending on whether improvement is summarized by trial-mean PER or phoneme-weighted session PER, these older sources outperform the previous-session source in a small minority of non-previous selections. This suggests that some target sessions may resemble older input-layer states rather than only drifting monotonically away from the previous day.

To inspect these non-previous selections directly at $K = 20$, we compare the geometry-selected source to the immediately previous source for each target session. Geometry selects the previous source in 26 sessions and an older non-previous source in 11 sessions. Among those 11 older selections, the older source achieves lower PER in 3 cases, the previous source remains better in 7 cases, and 1 case is tied. Thus geometry identifies plausible older-state candidates, but covariance distance alone is not sufficient as a deployment policy.

One possible explanation is that a day-specific input layer may represent a neural recording state rather than a purely chronological day. Changes in recording quality, electrode yield, thresholding, neural strategy, or user state could move a new session closer to an older feature geometry than to the immediately preceding session. This view is also consistent with evidence that local

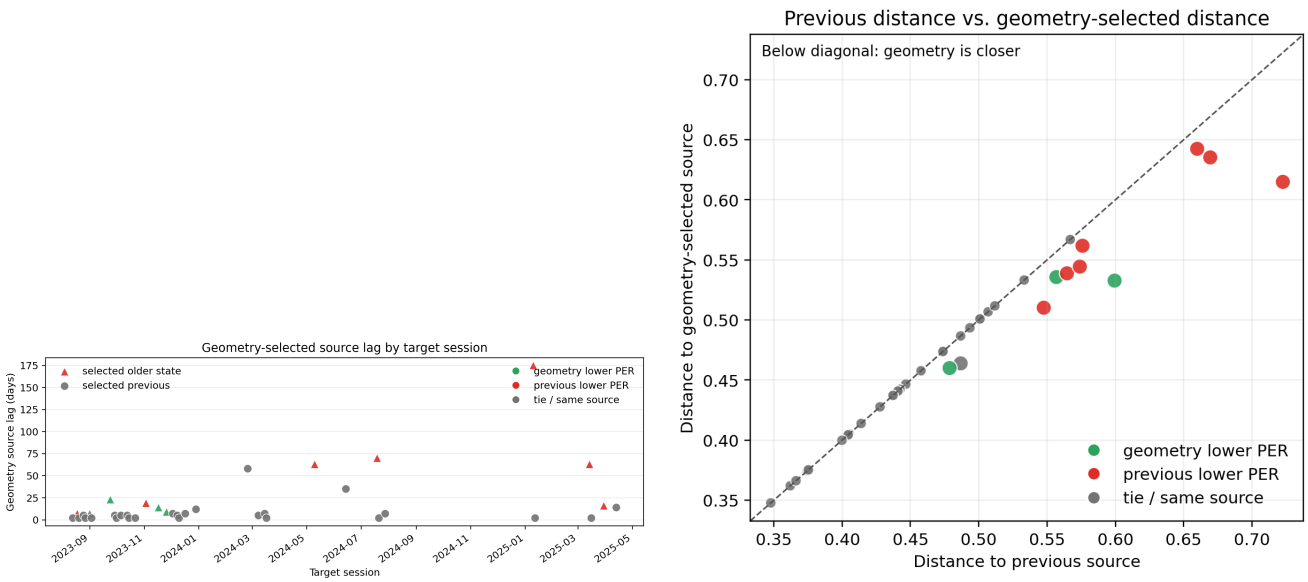

Figure 5: Geometry-selected older sources are meaningful candidates but not reliable standalone decisions at $K = 20$. Left: most target sessions select the immediately previous session, but several jump to older stored input states. Right: geometry-selected sources are closer in covariance distance by construction, but lower geometry distance does not always translate to lower PER. Green points indicate cases where the geometry-selected source outperforms previous; red points indicate cases where previous remains better.

Table 3: Evidence for non-recent source matches. These examples do not prove exact recurrence of neural states, but motivate maintaining a library of past input-layer states rather than only the most recent layer.

Data	Setting	Evidence	Interpretation
T15	K-shot	minority wins	override signal
T15	fingerprint	lower subspace angle	drift type
T12	diagnostic	08-13 → 06-21	53-day match
T12	diagnostic	08-25 → 08-13	older over previous

Table 4: K-shot input-state library-size ablation. PER is measured on remaining validation trials after using the first K trials to estimate target geometry. A bounded recent library nearly matches all-past selection, while previous-only recency remains the strongest aggregate baseline.

K	Previous	Last 3	Last 5	All past
5	13.09%	13.47%	13.55%	14.00%
10	12.84%	13.19%	13.19%	13.19%
20	12.21%	12.76%	12.74%	12.74%

cortical network activity can be modulated by action-perception state [Engelhard et al., 2019]. In this view, source selection is closer to memory retrieval over past input states than to choosing the most recent date.

We observe the same qualitative pattern in the T12 diagnostic-block geometry check. Using spike-power features, geometry selects an older non-previous session in 7/15 evaluable targets. Some selections jump substantially backward in time, for example 2022-08-13 selecting 2022-06-21, a 53-day lag, and 2022-08-25 selecting 2022-08-13 rather than the immediately preceding session.

The practical implication is that a longitudinal BCI should not keep only the latest input layer. A more robust system could maintain a small library of past input-layer states, use the previous session as the default, and invoke a recency-aware geometry gate only when the beginning-of-day neural geometry strongly suggests that an older state is a better match.

We therefore asked how large this past-state library needs to be. For each target session, we restricted K-shot geometry selection to the previous session, the last 3 sessions, the last 5 sessions, or all past sessions. The immediately previous session remains the best aggregate policy, but a small recent library is close to the all-past library. At $K = 20$, last-5 and all-past selection both reach 12.74% PER, while last-3 gives 12.76%. Expanding the nearest-source search to all past layers does not improve aggregate PER over recency under this simple distance rule; rather, it increases the need for a confidence gate before using older states.

5.4 Gating and Drift Fingerprints

This experiment asks why geometry is useful but hard to deploy: can it safely override recency, and when do older states actually help? Since geometry sometimes chooses older sources, we tested whether a simple distance-ratio gate can improve over recency: keep the previous source unless the geometry-selected source is at least an α fraction closer in covariance distance. This rule does not yet produce a meaningful improvement. At $\alpha = 0.9$, it ties previous at $K = 5$, is slightly worse at $K = 10$ (12.90%

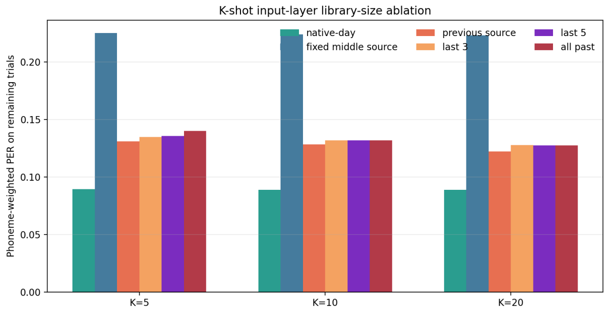

Figure 6: Library-size ablation. A small recent library approaches all-past geometry selection, but previous-session recency remains the strongest aggregate policy under the current covariance-nearest rule.

Table 5: Leave-one-session-out learned gate. The gate defaults to the previous source and learns a one-feature decision stump for when to override with the geometry-selected older source. It gives only a tiny gain at $K = 20$. Lower PER is better.

K	Previous	Geometry	Gate	Overrides
5	13.09%	14.00%	13.37%	6
10	12.84%	13.19%	13.18%	5
20	12.21%	12.74%	12.15%	4

vs 12.84%), and improves only marginally at $K = 20$ (12.18% vs 12.21%). At $\alpha = 1.0$, it approaches unconditional geometry selection and becomes worse. Geometry is therefore a useful candidate signal, but a deployable gate needs richer confidence features.

To test whether this failure is merely due to a poor hand-chosen threshold, we also train a tiny recency-aware gate. For each K, we fit a leave-one-session-out decision stump over geometry features such as distance ratio, absolute margin, source lag, and previous/source distances. This learned gate improves over previous-session recency only at $K = 20$, and only slightly (12.21% to 12.15% PER; Table 5). At $K = 5$ and $K = 10$, it overuses older sources and performs worse than previous. A one-dimensional learned gate is therefore still not enough; future gates need richer confidence signals.

The non-previous selections explain why the problem is difficult. Older sources beat the previous source in only a minority of non-previous geometry selections (3/11, 4/9, and 3/11 for $K = 5, 10, 20$ using phoneme-weighted session PER), and the Spearman correlation between geometry/previous distance ratio and older-source gain is weak ($\rho = 0.12, -0.02, 0.28$). Thus covariance geometry is useful for candidate retrieval, but not yet reliable as a standalone confidence measure.

To ask why older sources sometimes beat the previous session, we compare drift fingerprints for target–previous pairs and target–older-source pairs. This analysis is preliminary because older-source wins are rare, but it reveals a consistent pattern: wins are better explained by subspace geometry than by mean/scale shifts alone.

Table 6: Preliminary drift fingerprints for non-previous source choices. Negative principal-angle change means the older source is closer to the target subspace than the previous source.

K	Case	n	Δ principal angle
5	older wins	3	-1.37°
5	older loses	8	-0.06°
10	older wins	4	-1.93°
10	older loses	5	+1.02°
20	older wins	3	-2.91°
20	older loses	8	+0.55°

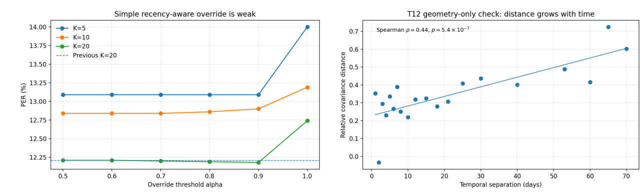

Figure 7: Left: a naive recency-aware geometry override does not substantially beat the previous-session baseline. Right: T12 diagnostic-block covariance distance increases with temporal separation, supporting the broader geometry framing outside T15.

When older sources win, they reduce the mean principal angle relative to the previous source by 1.37°, 1.93°, and 2.91° for $K = 5, 10, 20$. When older sources lose, this subspace improvement is absent or reverses (Table 6). This suggests that non-recent input states are useful when they better match the target neural subspace, supporting a drift-typing view of the adaptation ladder.

5.5 T12 Diagnostic Blocks Support the Geometry/Recency Framing

T12 diagnostic blocks provide a secondary geometry-only check. Using spike-power features, covariance distance increases with temporal separation across 120 past-source pairs (Spearman $\rho = 0.44$, $p = 5.4 \times 10^{-7}$). Geometry selects the immediately previous session in 8/15 target sessions and an older source in 7/15. Some selections jump far back in time, such as a 53-day lag. This pattern is not limited to spike power: threshold-crossing variants tx1–tx4 also show positive days-distance correlations ($\rho = 0.34$–0.47, all $p < 1.3 \times 10^{-4}$) and older-source selections in 6–9 of 15 targets. This supports the idea that recency is important but not sufficient to describe longitudinal neural geometry. Because we do not run the T12 decoder, this result is only supporting evidence, not decoder replication.

5.6 Inner-Speech Blocks Suggest Behavior States

As a geometry-only extension, we also analyzed interleaved verbal-behavior blocks from the T12/T15/T16/T17 inner-speech dataset [Kunz et al., 2025]. We did not train a decoder, evaluate PER,

Table 7: T12 diagnostic-block feature robustness. Across spike-power and threshold-crossing variants, covariance distance increases with temporal separation and geometry sometimes selects older non-previous sessions. This is a geometry-only supporting check.

Feature	ρ(days, cov.)	Previous	Older
spikePow	0.44	8/15	7/15
tx1	0.35	7/15	8/15
tx2	0.34	6/15	9/15
tx3	0.34	9/15	6/15
tx4	0.47	9/15	6/15

Table 8: Preliminary inner-speech behavior-geometry feasibility. This analysis is a supporting geometry-only check and does not evaluate decoding.

Feat./epoch	Acc.	Chance	B/W ratio
binnedTX / go	96.15%	20%	2.60
spikePow / go	84.62%	20%	2.87
binnedTX / delay	88.46%	20%	1.59

or attempt privacy-sensitive inner-speech decoding. Instead, within each participant, we grouped neural activity by speech behavior and split each behavior into two halves. We then asked whether covariance geometry retrieves the matching behavior. For binned threshold crossings during the go epoch, nearest-behavior accuracy was 25/26 (96.15%), compared with a 20% chance level, and between-behavior distances were 2.60 times within-behavior distances. Spike-power features and delay-epoch threshold crossings also remained well above chance (Table 8). These preliminary results suggest that the input-state view may extend beyond recording days: intended speech behavior is another axis of neural geometry.

6 Discussion

The main lesson is that cross-session adaptation should not be treated as a single fixed correction. The native T15 input layers are highly effective, and using the wrong input layer causes large decoder-facing degradation. But simple moment alignment does not emulate those layers, and learned adapters recover only part of the lost performance even when input-layer calibration is trained longer.

A small number of difficult sessions dominate worst-case PER, especially 2025-03-30 and 2025-03-14. A sensitivity check excluding the worst one or two sessions preserves the qualitative ordering: fixed non-native layers remain strongly degraded, moment matching remains ineffective, input-layer calibration improves PER only partially, and previous/geometry source policies remain the strongest low-cost choices. We therefore retain all sessions in the main analysis and treat these difficult days as stress cases rather than exclusions.

A practical interpretation is a beginning-of-day triage procedure. The system should first try the previous input state, because it is the strongest simple default when it is truly recent. If short-window geometry suggests that the new session resembles an older stored state, the system can retrieve that state. If no stored state appears reliable, lightweight residual calibration can update the input mapping while leaving the GRU backbone and output head frozen.

Table 9: Deployment interpretation of the adaptation ladder. The goal is not to apply every adaptation step every day, but to choose the lightest action that is supported by the beginning-of-day evidence.

Situation	Suggested action	Evidence in this audit
Recent previous	Reuse previous	strongest baseline
Fixed-source mismatch	Avoid fixed source	22.67–34.43% PER
Mean/scale shift only	Moment correction insufficient	$\sim$22.7% PER
Labeled K trials	Calibrate input	22.67% $\rightarrow$ 18.59% PER
Non-recent match	Retrieve older state	lower subspace mismatch
Behavior shift	Index behavior	inner-speech geometry

Table 9 summarizes the practical interpretation: the system should not perform the same update every day. Instead, it should start from the cheapest reliable action and escalate only when the beginning-of-day evidence suggests that the current input state is no longer reliable.

The most useful low-cost intervention is not a new adapter, but a better choice of existing input layer. A short unlabeled beginning-of-day window is enough to choose a past layer that recovers much of the fixed-source loss. Yet the strongest simple policy is often even simpler: use the previous session. This changes the adaptation ladder. Recency should be the default prior; geometry should be used as an additional signal for detecting when the previous session is not the right source. This also suggests a practical memory mechanism: instead of overwriting old session adapters, the system should retain a library of past input-layer states and learn when the current session has returned to one of them.

The failed distance-ratio override is also informative. It shows that geometry is not automatically a better policy than recency. The drift-fingerprint analysis suggests that the next gate should not rely only on covariance distance; subspace mismatch appears more informative for deciding when an older state is worth retrieving. A future gate should likely combine subspace features with distance margin, previous-source rank, temporal gap, session quality, decoder confidence, and perhaps hidden-state or logit-space diagnostics.

The library-size ablation further suggests that memory need not be unbounded. Under the current covariance-nearest rule, last-5 selection matches all-past selection at $K = 20$, while previous-only remains the best aggregate baseline. This points toward a compact state library with a conservative retrieval policy: keep a small set of recent or representative input states, default to recency, and use geometry only when multiple signals agree that an older state is likely to be better.

This framing also clarifies how the work relates to stronger decoder-backbone approaches. Pretrained speech models such as Wav2Vec2 may improve the representational power of future brain-to-text decoders, while our results address a different deployment problem: even with a fixed decoder backbone, the input mapping can drift across sessions and must be selected or adapted efficiently. Thus backbone improvements and input-state adaptation are complementary.

The inner-speech feasibility result suggests the same memory view may apply across behaviors. Future input-state libraries may need to index both recording state and intended speech mode, because a new neural geometry may reflect session drift, speech-mode shift, or both.

Deployment implications. The practical value of the adaptation ladder is that it avoids treating every new day as a full recalibration problem. A deployed system could begin with a small beginning-of-day window and ask a sequence of increasingly expensive questions. First, is the previous input state still recent and geometrically plausible? If so, the system can reuse it with no training. Second, does the new session resemble an older stored state more than the immediately previous one? If so, geometry can retrieve that state from a compact input-state library. Third, if stored states are not reliable and labels are available, a residual input calibration can update only the input transform while keeping the GRU backbone and output head fixed. This interpretation is deliberately conservative: geometry is not treated as a universal replacement for recency, and calibration is not treated as a reset. Instead, both are used as escalation steps when the cheaper prior state is no longer trustworthy.

Future deployment roadmap. The next step is to turn the adaptation ladder into an explicit online policy. Such a policy should not ask only whether geometry can beat recency. Instead, it should decide whether the current session is close enough to reuse the previous input state, whether an older stored state is a better geometric match, or whether labeled residual calibration is required. Our exploratory analyses suggest that this decision should combine temporal gap, subspace mismatch, source-library rank, decoder confidence, and session-quality features. This provides a practical path from an offline adaptation audit toward a deployable BCI calibration controller.

A second direction is to evaluate the same ladder with stronger decoding metrics and additional datasets. The present study reports greedy phoneme error rate, which isolates decoder-facing input mismatch without relying on a language-model stack. A full deployment study should also test word error rate, language-model decoding, and closed-loop or online adaptation. Similarly, the T12 and inner-speech results are intentionally geometry-only; future work should ask whether the same input-state and behavior-state structure improves actual decoding when suitable checkpoints and evaluation pipelines are available. A preliminary alignment direction is also worth treating cautiously: geometry-based feature transformations, such as covariance alignment, may introduce harms, so geometry is currently more reliable as a retrieval signal than as an automatic feature transformation. Another direction is to test whether the input-state library view extends beyond speech and motor decoding to cognitive or memory-related neural recordings. Recent human hippocampal and entorhinal single-neuron work shows that neural populations encode the temporal structure of experience [Tacikowski et al., 2024]. Such datasets could test whether geometry-guided state retrieval is specific to speech BCI drift, or whether it reflects a broader principle of reusable neural state organization.

7 Limitations

This study is a decoder-facing audit, not a new decoding architecture. Greedy PER is lighter than full WER and avoids the language-model/Redis evaluation stack. T12 is geometry-only and should not be interpreted as T12 PER replication. The inner-speech analysis is also geometry-only; it does not decode imagined speech or evaluate a communication system. The learned adapter and K-shot source-selection experiments simulate a beginning-of-day calibration window within the validation split and evaluate only on the remaining trials. They are intended to map adaptation rungs, not to optimize a final deployment benchmark. We focus the main claims on $K \leq 20$; larger calibration windows can be informative as feasibility checks, but in this validation split they consume a large fraction of the available session and are therefore not treated as minimal calibration.

8 Conclusion

For T15 brain-to-text decoding, fixed non-native input layers break the decoder, moment correction is too weak, and learned input adaptation improves PER only partially. The largest practical improvement comes from selecting a compatible past input layer. The immediately previous session is a strong default, while drift geometry is a measurable signal for deciding when that default may need to be overridden. This supports a recency-aware, geometry-guided adaptation ladder: start with the previous input state, measure drift geometry from a short unlabeled window, and move to stronger calibration only when the geometry indicates that simple source selection is not enough.

Ethical Statement

This work uses publicly released intracortical BCI datasets and reports only aggregate analyses. It does not attempt to infer private inner speech or deploy a decoder. Future extensions to inner-speech or cross-behavior state selection should include explicit intent-gating and privacy safeguards.

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